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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Miscellaneous – Lecture 02: Eigenvalues and Eigenvectors from a Data Sample

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Supplementary derivation lecture: **from raw sample data to covariance eigenvectors**

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

Agenda

Context

Raw Sample

Centering

Covariance

Eigenvalues

Eigenvectors

Interpretation

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

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By the end of this lecture, you should be able to:

- start from a small raw dataset and compute the mean vector,
- center the data and build the covariance matrix,
- solve for eigenvalues and eigenvectors by hand,
- interpret the largest eigenvector as the first PCA direction.

Why This Matters

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- In **PCA**, eigenvectors define the principal directions.
- In **LDA**, a related eigen-problem appears using scatter matrices.
- This lecture shows the mechanics on a tiny dataset where hand calculation is still realistic.

Raw Sample Data

Consider three two-dimensional samples:

$$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Goal: compute eigenvalues and eigenvectors **from this sample**, not from an already given covariance matrix.

Step 1: Compute the Mean

Mean vector:

$$\mu = \frac{1}{3}\left(\begin{bmatrix}2\\1\end{bmatrix} + \begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}0\\0\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}$$

Step 2: Center the Data

Subtract the mean from each sample:

$$x_1 - \mu = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_2 - \mu = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x_3 - \mu = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

So the centered data matrix is:

$$X_c = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Step 3: Build the Covariance Matrix

$$C = \frac{1}{n-1} X_c^\top X_c$$

Here $n = 3$, so:

$$X_c^\top X_c = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Therefore:

$$C = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

Step 4: Solve the Characteristic Equation

We solve:

$$\det(C - \lambda I) = 0$$

$$\det \begin{bmatrix} 1-\lambda & 0.5 \\ 0.5 & 1-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)^2 - 0.25 = 0$$

$$\lambda^2 - 2\lambda + 0.75 = 0$$

So the eigenvalues are:

$$\lambda_1 = 1.5, \qquad \lambda_2 = 0.5$$

Step 5: Solve for Eigenvectors

For $\lambda_1 = 1.5$:

$$C - 1.5I = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

Solving $(C - 1.5I)v = 0$ gives $y = x$, so:

$$v_1 \propto \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 0.5$:

$$v_2 \propto \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 6: Normalize the Eigenvectors

Normalized versions:

$$\hat{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \hat{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Since $\lambda_1 > \lambda_2$, $\hat{v}_1$ is the first principal direction for PCA.

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- The first principal direction is the diagonal line $x_2 = x_1$.
- The first eigenvalue is larger, so that direction captures more variance.
- The second direction is perpendicular and captures the remaining smaller spread.

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- Using raw data directly without centering it.
- Taking eigenvalues from the data matrix instead of the covariance matrix.
- Forgetting to normalize the eigenvectors.
- Forgetting that the largest eigenvalue corresponds to the first PCA component.

Summary

- Start from the raw sample, compute the mean, and center the data.
- Build the covariance matrix before solving the eigen-problem.
- Solve $\det(C - \lambda I) = 0$, then solve $(C - \lambda I)v = 0$.
- In PCA, the eigenvector of the largest eigenvalue becomes the first principal component.

Sources: Jolliffe and Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A*, 2016.;Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.;Hyvärinen and Oja, "Independent component analysis: algorithms and applications," *Neural Networks*, 2000.;Bishop, *Pattern Recognition and Machine Learning*, selected sections on PCA, LDA, and ICA.;Scikit-learn documentation, accessed 2026-04-18.

Exit Questions

1. Why do we center the data before computing covariance?
2. What matrix do we use for the eigenvalue problem in PCA?
3. Which eigenvector becomes PC1?
4. Why must eigenvectors usually be normalized?